Exact Product Capture and Context-Safe Weight Broadcast for Event-Driven Spiking Optical Flow

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Abstract—**Low firing activity does not automatically accelerate event-driven optical flow: finite storage, single-port state conflicts, and repeated weight delivery can dominate useful arithmetic. We present typed-source execution islands for a frozen binary-event optical-flow network. C1 combines exact repeated-product capture with finite single-port parent state and dead-write suppression. On 51.84 million source rows from ten `zurich_city_09_a` samples, its same-ledger cycle model reduces component time by 40.99% (1.6945×); a mapped 28-nm nine-SRAM component meets 3-ns setup/hold and passes 16,549 mapped-to-mapped Formality compare points. C2 shares Acc24 state across eight typed-signed sources. Across five directed workloads, against equal-bandwidth K1×8, K8 takes 1,913 versus 1,945 directed VCS cycles (1.0167×) and yields 4.5411× directed-throughput/logic-area with 77.61% less synthesized logic. Token-set broadcast then reuses weight delivery without reusing signed products. Across 1,920 fixed ep34 workloads spanning 40 samples and four DSEC sequences, same-port/cache RTL reduces post-load VCS cycles from 12,522,876 to 5,124,365 (2.4438×; 59.08%) and weight requests by 64.25%; a matched logic-only DC ablation adds 0.0118% area. Matched hold and power remain open. C3 closes exact fixed-T=10 temporal service at 3 ns. An operator-local INT8 bridge matches 1,280 integer-oracle probes and stays within Acc24; a separately evaluated mixed-precision deployment candidate passes a predefined AEE compatibility gate on an 825-frame local DSEC validation set. We keep model, RTL, and physical evidence separate and make no whole-network speedup claim.**

I. Introduction

Event cameras make inactivity visible, but sparse arithmetic is only one term in execution time. In our optical-flow deployment, nonzero sources still trigger parent-state traffic, accumulator ownership, and weight-row fetches. Consequently, a design that reports only operation sparsity can be slower than an equal-resource zero-skipping baseline. Prosperity [1] and Phi [2] demonstrate that useful SNN speedups come from executable reuse or patterns, not firing rate alone; FireFly-T [3] shows the value of multi-spike service. Our problem differs in its finite 240-KiB-class working set, polarity-bearing typed sources, and mixed fixed-T neuron phases.

We make two architectural contributions. (1) C1 captures exact repeated products subject to a finite-capacity, single-1RW parent-store contract. (2) C2 shares Acc24 and endpoint state across eight typed sources and embeds token-set broadcast (TSBG), which reuses a weight row while preserving independent signed products and accumulator contexts. C3 provides exact fixed-T=10 temporal service as network-coverage evidence rather than a third performance claim. The checkpoint is an evaluation workload rather than a parallel algorithmic novelty claim.

II. Workload and Execution Contract

The frozen Motion C12 ep34 checkpoint obtains 1.199514 AEE on an 825-frame local DSEC validation set and has a 5.6709% global firing rate. Of 105 configured ATLIF wrappers, 12 `sn2_q` wrappers are runtime-bypassed and never called, so the capture invokes 93 (48 T=2, 45 T=10). A separate graph audit finds that 12 invoked `attn_sn` return values have no consumer under fixed normal inference, leaving 81 graph-live services under that fixed call graph (36 T=2, 45 T=10). All 93 captured ATLIF outputs are binary. Signed source codes are therefore a downstream execution-protocol property—for example polarity and correction traffic—not evidence that the captured ATLIF tensors are analog. The frozen software path uses Motion-XOR scoring with α=0.125 and K as its value carrier, but its configuration has hardware quantization disabled. Only the separately evaluated deployment candidate enables Q7 round-to-nearest scores, a next-power-of-two integer Shiftmax row sum, and Q1.7 gates. Attention is not promoted to a performance contribution, nor are its fractional gate values called powers of two.

Each result is assigned one evidence class. A cycle-model result replays the same trace and charged resources in software. A VCS result proves directed RTL function/protocol properties. DC/PT/Formality results anchor area, timing, and equivalence. Only a result that closes all three classes on one workload may be called an RTL speedup; the present paper instead reports the three classes side by side.

The mapping is driven by three exact workload invariants rather than a claim that every layer has the same sparsity. Bottleneck convolution exposes repeated products, but only while parent and destination state remain resident; this defines C1. FC tokens can share a weight-row identity while retaining different signs, destinations, and accumulators; C2 therefore reuses delivery, not products, through TSBG. Fixed-T=10 neurons require a deterministic state and commit order; C3 preserves that order. Attention and decoder traces remain part of the frozen workload identity but are not promoted to additional performance contributions.

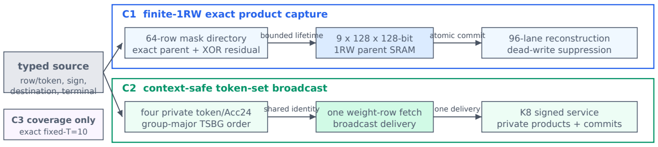

Fig. 1. Two exact reuse objects under one typed-source protocol. C1 retains only legal live parents in a finite 1RW store; C2/TSBG broadcasts one delivered weight row while keeping products and Acc24 contexts private. C3 is coverage, not a third performance claim; the branches are not a measured monolithic top.

TABLE I
C1 SAME-LEDGER CAPTURE LADDER (51.84 M SOURCE ROWS).

Execution point	Cycles (M)	vs. zero	Boundary
Strong zero	648.741	1.000×	baseline
Same-coordinate bit	646.619	1.003×	baseline
C1 finite 1RW	382.849	1.6945×	cycle model
Concurrent-access	341.058	1.902×	ceiling only

III. MICROARCHITECTURE

A. C1: Capacity-Constrained Product Capture

C1 first builds a ping-pong directory for each 64-row, 16-source mask task. A row may select only an earlier exact-subset row as its parent; it then issues the XOR residual and reconstructs the 96-lane signed product row by adding the parent. Stable population-count ordering and a live-parent bitmap make this relation deterministic and identify final values that no later row can observe. Such dead writes are elided exactly.

The execution side combines one earliest-parent lookahead, a two-entry reserved response queue, and forwarding under a single-1RW deadline-aware arbiter. Final psum write and row completion are atomic, so backpressure cannot expose a partially reconstructed row. The scheduler uses nine 128×128 1RW SRAM macros and never assumes the ideal simultaneous access that produced the higher but unphysicalized reuse ceiling. This is the object difference from an unbounded product-sparsity opportunity: reuse must survive capacity, port, parent lifetime, and completion constraints.

For parent mask $\mathcal{M}_p \subset \mathcal{M}_r$, exactness follows from the disjoint residual rather than value prediction:

$$\mathbf{P}_r = \mathbf{P}_p + \sum_{s \in \mathcal{M}_r \setminus \mathcal{M}_p} \mathbf{p}_s. \tag{1}$$

Table I exposes the baseline ladder instead of attributing all inactivity to product capture. Plain bit skipping barely changes time on this ledger; exact parent reuse creates the material reduction. The remaining 12.25% gap from C1 to the concurrent-access ceiling is the price of the physical single-port contract, not an omitted claimed speedup.

B. C2: Typed K8 and Token-Set Broadcast

The K8 fabric accepts at most eight nonzero typed sources and shares the Acc24 array, endpoint, and control rather than replicating eight K1 engines. A fair baseline is therefore K1×8 at equal source bandwidth, not one K1 lane. TSBG changes only weight delivery. Four token contexts expose a common source-group row; one fetched row is broadcast to four independent signed products and Acc24 contexts. It never reuses a product because equal source indices can carry different signs or destinations. Both ordinary and TSBG modes use the same 192-bit row-live representation and hierarchical priority encoder, avoiding division/remainder hardware that would bias the physical comparison. Bundled event delivery and mini-batch Gustavson execution are established in ELSA [4], while SpikeX [5] reuses weights across time and space. Our narrower contribution is their mapping to a polarity-bearing source stream with four private destination/Acc24 contexts under the same finite row cache and public-port contract. For a common weight row W_g, TSBG shares only its delivery:

$$\mathrm{fetch}(W_g) = 1, \quad A_c \leftarrow A_c + v_c W_g, \quad (v_c, d_c, A_c) \text{ private.} \tag{2}$$

C. C3: Exact Fixed-T Coverage

C3 schedules the deployed T=10 ATLIF recurrence and preserves its integer state and commit order. It is included to close the temporal service boundary, not to claim a standalone speedup.

All performance mechanisms have an exact fallback. If C1 finds no resident legal parent, it selects the empty parent and issues the original source row. If token contexts disagree on a weight-row identity, C2 emits separate fetches; sign, destination, and Acc24 state are never coalesced. Backpressure changes only service order, because the typed-source terminal bit and atomic commit keep ownership explicit. Thus sparsity affects delivered work without changing the operator result.

IV. EVALUATION

All physical results use a commercial 28-nm flow at 3.0 ns. The reported points are prelayout, use ideal clocks and ZeroWireload, and have no extracted parasitics; C1 area includes nine SRAM macros, whereas C2 and C3 are logic-only. The component comparison in Table II deliberately keeps cycle, area, and proof scope visible. C1 model cycles replay ten `zurich_city_09_a` ep34 samples and 51.84 million source rows; per-sample ratios span 1.661–1.723×, with a 1.6946× geometric mean. C2 VCS cycles use five frozen directed workloads. TSBG's four-sequence VCS distribution uses all 40 captured samples, all 12 FC1 layers, the four FC2 layers at or below G48, and first/middle/last aligned B4 quartets; the 1,920-workload selection is fixed independently of performance. The sealed population combines 1,917 inherited logs with three successor logs from the same compiled simulation image. M2053 is not promoted as a successful result; only its individually valid logs are inherited with explicit lineage. A broader software screening uses 960 FC1/FC2 pairs from the same samples and sequences. Task accuracy binds the frozen checkpoint to a deployment subset on the same population and metric aggregation. Because the historical baseline and candidate use different GPU backend flags, their difference is baseline-referenced rather than a single-factor causal estimate.

We do not claim full-network INT8 or a whole-network RTL speedup.

AEE (average endpoint error) is the preselected compatibility-gate metric. AAE is the legacy 2-D angular error; AAE-Bench. is the Middlebury/Barron 3-D angular error; DSEC-Fl is the percentage with endpoint error above 3 pixels and 5% of ground-truth flow magnitude.

Table III uses the same 825 frames, 18 sequences, and 48,152,523 valid pixels. The candidate applies hardware-order attention to all 12 blocks and dyadic-INT8 QDQ to four C1 Conv3x3 and four decoder ConvTranspose weights; every transformed weight is reached once per frame. Other operators remain at checkpoint precision. The historical baseline enabled TF32/cuDNN benchmarking whereas the candidate disables both; therefore the table demonstrates accuracy-gate compatibility, not a causal accuracy improvement from quantization. Aggregate AEE falls by 0.002147, but the three auxiliary errors in Table III rise and 10 of 18 sequences regress in AEE; we claim only frozen-population gate compatibility, not universal improvement.

The C1 mapped component includes its nine SRAM leaves. It dissipates 29.08 mW and 22.07 nJ in a separately bounded 64-row, 253-cycle directed window. This is a mixed-corner, prelayout estimate without SPEF, not energy per frame. C2 K8 and K1×8 meet setup in fresh matched synthesis; independent matched hold and power remain outside the admitted row.

For C1, a separately sealed ep34 tile uses the first 64 real support masks of the cycle ledger. VCS matches 196 accepted issues, 58 parent edges, 31 dead-write elisions, 54/33 macro reads/writes, four forwards, six deadline holds, 14 stalls, and 64 commits/completions. Its lane values are deterministic synthetic signed-12-bit data with zero prior psum; this calibrates modeled event counts and directed arithmetic, not the 1.6945× cycle ratio itself.

For numerical binding, all eight retained Conv3x3/ConvTranspose weights are quantized per output channel to narrow-range signed INT8 with a power-of-two scale. An independently checked export has zero pre-clip violations; the largest absolute-code accumulator bounds are 200,219 for C1 and 87,136 for the decoder, both below Acc24. The subsequent operator-local replay covers 160 full calls and 3,271,680,000 output elements. Against FP32 it gives 0.009106 MAE, 0.012407 RMSE, 0.239204 maximum absolute error, and 0.9999789 cosine similarity. All 1,280 independently sampled Python-integer probes match; the full observed final-accumulator range is $[-29{,}680, 27{,}619]$, and the formal worst-case prefix bound is 200,219, both within signed Acc24. These are operator-local numerical results, not task AEE, downstream-neuron equivalence, or a hardware speedup. Table III supplies the separate task-level accuracy binding without promoting this operator-local bridge to a full-network integer implementation. As a C2 scheduling study, TSBG's full-capture software replay reduces serialized modeled cycles from 29.41 billion to 11.61 billion and weight traffic by 65.20%[model], with a minimum per-sequence ratio of 2.517×. A separate VCS distribution fixes, before measuring performance, all 40 samples, all 12 FC1 layers and the four FC2 layers whose inputs fit the same G48 engine, and first/middle/last aligned B4 token quartets. Across these 1,920 workloads, ordinary LRU4 takes 12,522,876 post-load execute cycles versus 5,124,365 for TSBG-B4 (2.4438×); scalar weight requests fall from 8,774,304 to 3,136,608 (−64.25%). FC1/FC2 weighted ratios are 2.5466/1.9680×; the four sequence ratios span 2.3814–2.5458×. The population retains 286 all-zero workloads; 1,343 workloads improve, 570 tie, and seven are slower, with a median of 1.8373× and worst nonempty ratio of 0.9935×. The fixture supplies real ep34 activity/sign descriptors; all naturally nonzero codes in this measured population are +1. Deterministic INT8 weights exercise arithmetic but do not affect scheduling. The same runs check bank stalls/reordering and stale responses; a subsequent synthetic recovery phase checks signed products, reset, and the −128 corner. This is an RTL component microbenchmark, not full-FC or system speedup. The physical comparison elaborates the same B4 RTL with only `SCHEDULE_MODE=0/1` changed; row-live representation, cache capacity, public ports, and synthesis constraints are otherwise identical. In that matched logic-only ablation, ordinary LRU4 and TSBG-B4 occupy 249,710 and 249,740 μm^2, respectively, a 0.0118% increment, and both meet 3-ns setup. Both have a diagnostic hold slack of −16.4 ps. State is implemented with standard cells in this experiment; power, SRAM-macro cost, and hold closure remain open.

Table IV reports the full 40-sample G48 VCS distribution; it is not promoted to full-network speedup.

V. Related Work and Comparison Discipline

Prosperity reports average accelerator speedup against PTB and separately isolates product over bit sparsity [1]; the latter is the relevant mechanism-level comparison to C1. Phi chooses its storage point by DSE and compares its complete hierarchical-pattern architecture with published baselines [2]. We adopt the useful evaluation pattern—cycle simulation plus RTL/physical anchors and a baseline ladder—without relabeling their numbers as ours. ELSA and SpikeX establish bundle/Gustavson and cross-window weight-reuse priors [4], [5]; TSBG is therefore claimed only as a typed-signed, finite-context delivery specialization inside C2. FireFly-T's multi-spike decoder motivates an equal-service comparison for C2 [3]. Lossy pruning is outside the scope of this exact component paper.

SpinalFlow sorts compressed spike streams to preserve useful state locality, while SNE makes convolutional work proportional to the incoming event stream [6], [7]. They establish event-driven execution, but do not remove C1's repeated products under a finite 1RW parent lifetime or C2's repeated weight delivery across private signed contexts. A recent 28-nm event-based optical-flow chip jointly reports operations, external-memory traffic, latency, energy, and AEE [8]. We follow that separation of metrics, but do not numerically rank its silicon measurements against our prelayout component anchors.

TABLE II
Admitted component evidence; "open" claims are not inferred.

Line	Performance	Physical anchor	Evidence boundary
C1	1.6945×; −40.99% time[model]	166,514 μm^2; PT setup/hold +27.9/+1.8 ps; 16,549 mapped-to-mapped compare points pass	One-sequence model ratio plus a separate mapped physical anchor; one real-mask tile is event-count calibrated in VCS
C2	1,913 vs 1,945 directed cycles (1.0167×)[VCS]; 4.5411× directed-VCS throughput/DC logic-cell area	logic-only 131,086 vs 585,479 μm^2; −77.61%	K8 vs equal-bandwidth K1×8 in one admitted campaign; matched hold/power open
C2/TSBG	12,522,876 vs 5,124,365 execute cycles (2.4438×)[VCS]; time −59.08%; weight requests −64.25%	matched logic-only 249,710 vs 249,740 μm^2; +0.0118%; setup met	1,920 fixed ep34 real-activity workloads; 40 samples, four sequences, same G48 engine; identical 383-cycle preloads excluded; no full-FC/system claim
C3	17 cycles/tile	63,756 μm^2[DC/PT]; DC/PT setup+hold met; 11,180 compare points pass	Exact service; Formality compares pre- and post-hold-repair mapped netlists, not RTL-to-gate; no speedup/energy claim

TABLE III
Frozen-population local DSEC accuracy against a historical, backend-mismatched baseline (825 frames; lower is better).

Metric	Baseline	Candidate	Δ
AEE	1.199514	1.197367	−0.002147
AAE	5.400641	5.412808	+0.012167
AAE-Bench.	5.106363	5.121619	+0.015256
DSEC-Fl	5.313360	5.328834	+0.015475

TABLE IV
TSBG-B4 full 40-sample G48 VCS distribution.

DSEC trace	Ordinary cyc.	TSBG cyc.	Speedup
interlaken_01_a	3,120,326	1,225,652	2.5458×
thun_01_b	2,921,236	1,182,417	2.4706×
zurich_city_09_a	3,204,120	1,345,486	2.3814×
zurich_city_12_a	3,277,194	1,370,810	2.3907×
aggregate	**12,522,876**	**5,124,365**	**2.4438×**

For a co-designed network, a published chip rarely supports the identical graph. We therefore compare at three levels: same-network component baselines, iso-workload public-artifact opportunities, and published technology-normalized metrics. Results from different levels are never multiplied.

A. Limitations

The C1 ratio is a cycle-model result calibrated by one real-mask RTL tile; its mapped nine-parent-SRAM component is a separate physical anchor, not a full integration of the 240-KiB-class common-charge ledger. C2's area-efficiency comparison is logic-only. TSBG's RTL cycle result covers 1,920 fixed B4 workloads but excludes eight FC2 layers above G48 and the full token population; its full-capture external weight-service time remains modeled, and seven nonempty microbenchmarks are marginally slower, worst 0.9935×. The reported VCS population is a double-sealed 1,917+3 same-simulation-image cross-attempt result; its failed parent attempt remains non-citable. We leave matched-macro C2 hold/power, full-network FPS/energy, and monolithic island integration open instead of deriving them from component ratios.

VI. Conclusion

The principal hardware result is not firing sparsity itself, but its executable capture under finite storage and service constraints. C1 provides bounded exact product reuse; C2 turns typed multi-source service into area efficiency and uses TSBG to suppress repeated weight delivery; C3 closes exact temporal service. Separating model, VCS, and physical evidence yields a compact ISCAS component paper without an unsupported whole-network claim.

TABLE V
Mechanism boundary; prior numbers are not relabeled as ours.

Work	Prior object	Boundary retained here
Prosperity [1]	repeated products	C1 adds finite single-1RW parent lifetime, exact residual reconstruction, and atomic completion
Phi [2]	hierarchical patterns	no substitution for C1 dynamic parents or C2 signed contexts
FireFly-T [3]	multi-spike service	motivates equal-bandwidth K8, not a one-lane baseline
ELSA [4]/ SpikeX [5]	bundle/weight reuse	TSBG preserves private sign, destination, and Acc24 state
SpinalFlow [6]/ SNE [7]	event-stream execution	do not subsume finite-parent product reuse or typed cross-token delivery
CICC optical flow [8]	silicon system metrics	metric template only; no cross-platform speedup against prelayout islands
This work	exact parent + typed delivery	same-ledger model, VCS, and 28-nm islands; no system-speedup claim

References

[1] C. Wei, C. Guo, F. Cheng, S. Li, H. Yang, H. Li, and Y. Chen, "Prosperity: Accelerating spiking neural networks via product sparsity," in *Proc. IEEE HPCA*, 2025.

[2] C. Wei, B. Duan, C. Guo, J. Zhang, Q. Song, H. Li, and Y. Chen, "Phi: Leveraging pattern-based hierarchical sparsity for high-efficiency spiking neural networks," in *Proc. ACM/IEEE ISCA*, 2025.

[3] T. Li, J. Li, G. Shen, D. Zhao, Q. Zhang, and Y. Zeng, "Firefly-t: High-throughput sparsity exploitation for spiking transformer acceleration with dual-engine overlay architecture," *IEEE Transactions on Computers*, vol. 75, no. 6, pp. 2185–2199, 2026.

[4] K. You, C. Nie, L. J. Yan, Z. Wei, C. Zou, Z. Xu, Y. Feng, H. Jiang, and Z. He, "Elsa: An elastic snn inference architecture for efficient neuromorphic computing," in *Proc. ACM/IEEE ISCA*, 2026, pp. 2550–2566.

[5] B. Xu, R. Boone, and P. Li, "Spikex: Exploring accelerator architecture and network-hardware co-optimization for sparse spiking neural networks," *arXiv preprint arXiv:2505.12292*, 2025.

[6] S. Narayanan, K. Taht, R. Balasubramonian, E. Giacomin, and P.-E. Gaillardon, "Spinalflow: An architecture and dataflow tailored for spiking neural networks," in *Proc. ACM/IEEE ISCA*, 2020, pp. 349–362.

[7] A. D. Mauro, A. S. Prasad, Z. Huang, M. Spallanzani, F. Conti, and L. Benini, "Sne: An energy-proportional digital accelerator for sparse event-based convolutions," in *Proc. DATE*, 2022, pp. 825–830.

[8] T. Zhang, Y. Zhong, Y. Yang, J. Ruan, Y. Gao, Z. Zhang, and Y. Wang, "A 28-nm optical flow estimation accelerator with redundancy speculation, bit-width-aware compression and similarity detection," in *Proc. IEEE CICC*, 2026.